

# Magic of cosmological particle production and the de Sitter horizon

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## Abstract

Non-stabilizerness (“magic”) has been proposed as the quantum-information resource behind gravitational back-reaction, but that program lives almost entirely in anti-de Sitter and Sachdev–Ye–Kitaev (SYK) settings. We ask whether magic is computable, and behaves, in two settings closer to our own universe: cosmological particle production and the de Sitter horizon. Two results, both in deliberately conservative toy models. (i) Cosmological pair creation sources magic *by resource theory*: a free bosonic two-mode-squeezed out-state is Gaussian and hence Wigner-positive (magic-free), with any finite-qubit stabilizer entropy an encoding artifact; only genuine non-Gaussianity (interactions) turns on Wigner negativity. The fermionic out-state, by contrast, has exact qubit non-local magic  $-\log_2(4P^2 - 6P + 3)$  that is non-monotonic in the occupation, and on the sudden de Sitter→radiation transition this fermionic magic is strongly suppressed unless the transition is violent—so cosmological fermion magic is a probe of the violence of reheating. (ii) The de Sitter horizon, via its conjectured double-scaled-SYK dual, shows *no* magic signature at the static level, but a robust one in the dynamics: the stabilizer entropy of the time-evolved thermofield double grows about twice as fast for a chaotic Hamiltonian as for a free one, and the chaotic-free late-time gap grows with system size (+0.61, +1.03, +1.37 at  $N = 8, 10, 12$ ; positive under a same-spectrum control for  $N \geq 10$ ), the de Sitter analog of the magic↔area link. We state throughout what is a theorem, what is a conservative proxy, and where an initially over-stated claim was narrowed by adversarial cross-check.

## 1 Introduction

Entanglement entropy fixes the Ryu–Takayanagi *area*, but an area on a fixed background does not gravitate; producing state-dependent geometry and back-reaction requires a further resource, and recent work identifies it with non-stabilizerness, or “magic” [1–3]. That identification has been developed in AdS holographic codes and in SYK. Here we ask the two natural “our universe” questions it invites: does cosmological particle production generate magic (Section 2), and is the de Sitter horizon magical (Section 3)?

Both are addressed by exact computation in small toy models, reusing a Fast-Walsh–Hadamard stabilizer-entropy engine and standard SYK machinery. The organizing principle that emerges is a *resource-theory* one: in the continuous-variable description magic is Wigner

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negativity, and free (Gaussian) sectors are magic-free by Hudson’s theorem, so any magic must come from either the genuinely discrete (fermionic) sector or from interactions and chaos.

## 2 Magic of the expanding vacuum

An expanding background turns the in-vacuum into an out-state of created pairs of opposite comoving momentum—a two-mode-squeezed (boson) or single Pauli-blocked pair (fermion) per mode. We compute the magic of that out-state.

### 2.1 Boson: a theorem-anchored null, and the non-Gaussian turn-on

For a free scalar mode pair, cosmological particle production gives the two-mode squeezed vacuum

$$|\text{TMSV}(r)\rangle = \sqrt{1-\lambda^2} \sum_{n=0}^{\infty} \lambda^n |n_k, n_{-k}\rangle, \quad \lambda = \tanh r. \quad (1)$$

This is a pure Gaussian continuous-variable state. By Hudson’s theorem, and equivalently by the continuous-variable Gottesman–Knill simulation criterion [4, 5], its Wigner function is nonnegative for every squeezing parameter  $r$ . Its physical CV mana,

$$\mathcal{M}_{\text{CV}} = \log_2 \int dq dp |W(q, p)|, \quad (2)$$

is therefore exactly zero. Free bosonic inflation can create large entanglement and large occupation, but not Wigner negativity; this agrees with the recent “Inflation is Not Magic” analysis [7].

The important guardrail is that a finite Fock cutoff followed by a binary encoding is not the same resource theory. Truncating  $|\text{TMSV}\rangle$  to  $2^q$  oscillator levels and mapping the Fock labels to qubits can produce a nonzero qubit Pauli stabilizer entropy, but that number depends on the cutoff and on the arbitrary relabeling of Fock states once the encoding is large enough to admit non-affine permutations. The same underlying CV state still has  $\mathcal{M}_{\text{CV}} = 0$ . Thus the free-boson result is not a small qubit signal; it is a theorem-level null in the invariant CV phase-space resource theory.

Interactions supply the missing resource. As a minimal executable proxy for primordial non-Gaussianity generated by interactions, we used a cubic-phase deformation of a Gaussian wavepacket,

$$\begin{aligned} \psi_{\sigma, \gamma}(x) &= \sigma^{-1/2} \pi^{-1/4} \\ &\times \exp \left[ -\frac{1}{2}(x/\sigma)^2 + \frac{i\gamma}{3}(x/\sigma)^3 \right]. \end{aligned} \quad (3)$$

At  $\gamma = 0$  the direct phase-space quadrature gives zero negative volume. For  $\gamma > 0$ , Wigner-negative interference fringes appear: in our grid calculation the negative volume grows from  $6.9 \times 10^{-4}$  at  $\gamma = 0.24$  to  $7.4 \times 10^{-3}$  at  $\gamma = 0.40$  and  $3.3 \times 10^{-2}$  at  $\gamma = 0.70$ . A toy ultra-slow-roll proxy  $\gamma_{\text{eff}} = \gamma_0 e^{2\Delta N}$  reproduces the expected rapid growth once the non-Gaussianity becomes appreciable, matching the qualitative mechanism of (author?) [6]. The claim is deliberately narrow: this is not a reproduction of their full inflationary EFT calculation, but it isolates the resource-theory transition. Linear squeezing alone remains Gaussian and Wigner-positive; genuine non-Gaussianity is what turns bosonic cosmological magic on.

## 2.2 Fermion: exact qubit magic and the violence of reheating

Pauli exclusion makes the fermionic out-state a single pair per mode. Under the Jordan–Wigner map—which is exact because fermionic Fock space is two-dimensional per mode—mode  $k$  and its partner form the two-qubit state  $\alpha|00\rangle + \beta|11\rangle$  with  $|\beta|^2$  the occupation. Its qubit non-local magic (the stabilizer Rényi entropy minimized over local unitaries) has the closed form

$$\mathcal{M}(\beta) = -\log_2(4P^2 - 6P + 3), \quad P = |\alpha|^4 + |\beta|^4, \quad (4)$$

verified to machine precision against a direct local-unitary minimization. It is *non-monotonic*: it vanishes at the product limit ( $|\beta|^2 = 0$ ) and at the maximally-mixed “Bell” point ( $|\beta|^2 = \frac{1}{2}$ ), and peaks at  $\mathcal{M} = \log_2 \frac{4}{3} \approx 0.415$  where  $|\beta|^2 = \frac{1}{2}(1 \mp 1/\sqrt{2})$ . Cosmological creation thus writes magic into an intermediate band of the occupation.

The physically sharp statement rides the sudden de Sitter→radiation transition whose scalar Bogoliubov coefficients are analytic [8],  $|\beta_k|^2 = 1/(4x^4)$  with  $x = k|\eta_e|$ . The boson carries *no* magic across this whole spectrum (Gaussian, above). A massless fermion is conformally invariant in a conformally-flat FRW universe and is not produced at all—an exact null. A massive fermion is produced, but for the paper’s smooth ( $C^1$ ) transition the production is strongly suppressed (peak occupation  $\sim 3 \times 10^{-4}$ ): the mass term  $M = m a(\eta)$  is  $C^1$  and never couples to the  $a''/a$  jump that drives boson production. Only a genuinely violent (sudden,  $a'$ -discontinuous) transition, for which the sudden-approximation occupation  $n_k = \frac{1}{2}(1 - x/\sqrt{x^2 + \mu^2})$  runs from  $\frac{1}{2}$  in the infrared to 0 in the ultraviolet, drives the occupation through the magic band. Hence

*cosmological fermion magic is a probe of the violence of reheating:*

negligible for a smooth transition, band-level only for a sudden one, and zero for a massless (conformal) fermion. The boson/fermion asymmetry has a one-line root: fermionic Fock space is two-dimensional per mode, so its qubit magic is physical, whereas bosonic Fock space is infinite-dimensional and any qubit magic is a truncation artifact.

## 3 Is the de Sitter horizon magical?

The de Sitter static patch is conjecturally dual to double-scaled SYK [9, 10], giving a computable stand-in for the chaotic finite horizon Hilbert space; the honest dictionary is high-temperature and degree-of-freedom based, so we test scaling with a horizon-dof proxy rather than a temperature monotone. As a conservative executable proxy we use ordinary SYK<sub>4</sub> (chaotic) versus SYK<sub>2</sub> (free).

**The static level is a null.** At a fixed time, magic does not separate chaotic from free: the thermal-pure-quantum stabilizer entropy is extensive but barely differs between SYK<sub>4</sub> and SYK<sub>2</sub>; the twin-controlled thermal-state magic even runs backwards (chaos flattens the spectrum toward maximal mixedness); and the infinite-temperature doubled thermofield double  $\sum_E |E\rangle_L |E^*\rangle_R = \sum_i |i\rangle_L |i\rangle_R$  is exactly the Bell state, magic-free independent of the Hamiltonian. This null is confirmed across multiple measures.

**The dynamics reveals a signature.** The discriminator is the stabilizer entropy of the time-evolved thermofield double  $|\text{TFD}(\beta, t)\rangle \propto \sum_n e^{-\beta E_n/2} e^{-iE_n t} |n\rangle_L |n\rangle_R$ , whose magic dynamics is expected to track the spectral form factor [11]. Two discriminators, both with error bars over independent disorder realizations:

- **Scrambling rise.** Chaotic SYK<sub>4</sub> thermofield-double magic grows about twice as fast as free SYK<sub>2</sub> under evolution; the rise ratio is 2.05, 2.30, 2.55, 2.17 at  $N = 6, 8, 10, 12$ —robust across size.
- **Extensive plateau gap.** The late-time SYK<sub>4</sub>–SYK<sub>2</sub> magic gap is positive and grows with the number of degrees of freedom, +0.61, +1.03, +1.37 at  $N = 8, 10, 12$  (tight seeds at  $N = 12$ ,  $+1.37 \pm 0.07$ ). Under a same-spectrum diagonal-basis control that removes the eigenbasis contribution the gap is  $-0.14, +0.75, +0.94$  at  $N = 8, 10, 12$ : positive and growing for  $N \geq 10$ .

Strikingly, at a fixed time the *free* system carries more magic (the static gap is negative); only under evolution does the chaotic system reveal its larger magic capacity. The growth of the gap with the horizon-dof count is the de Sitter analog of the magic↔area link [1], seen dynamically where the static snapshot is flat.

## 4 Scope, and a note on method

Every model here is a toy: the fermion result is a two-level Dirac mode model, and the de Sitter results use a conservative SYK<sub>4</sub>/SYK<sub>2</sub> proxy for double-scaled SYK at finite  $N \leq 12$ , not the full constrained double-scaled construction. The claims are correspondingly bounded: the boson null is a theorem, the fermion reheating-violence statement is a mechanism in a toy transition, and the horizon-magic signature is a scrambling-sensitive diagnostic whose extensive scaling is firmed for  $N \geq 10$  but wants larger  $N$  / full DSSYK to become a theorem.

We record one methodological point deliberately. The extensive plateau gap was first reported from a single disorder seed as a clean trend from small  $N$ ; an independent-seed error-bar cross-check showed the smallest- $N$  point was a fluctuation (an actual sign crossover), and the claim was narrowed to its firmed form above. The correction, and the two agents’ mutually adversarial cross-checks that produced it, are part of the result.

## References

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